

Integrating Chaotic Dynamics with Time–Frequency EEG Features for Enhanced Epileptic Seizure Detection

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ABSTRACT

Epilepsy is a neurological disorder characterized by recurrent seizures resulting from abnormal brain electrical activity. Accurate seizure detection is essential for clinical diagnosis and continuous monitoring; however, conventional electroencephalography (EEG) analysis often fails to capture the nonlinear and time-varying dynamics associated with seizure onset. This study proposes an automated EEG-based seizure detection framework that integrates time-domain, frequency-domain, and nonlinear chaotic features to enhance classification performance. EEG recordings from the CHB-MIT scalp EEG database were processed using a four-stage pipeline consisting of band-pass filtering, power-line interference removal, wavelet-based denoising, and artifact suppression. Features were extracted from multiple channels, including nonlinear descriptors such as the Largest Lyapunov Exponent. Principal Component Analysis (PCA) and SMOTE-based class balancing were applied to improve efficiency and robustness. Three machine-learning classifiers—Random Forest, Extreme Gradient Boosting (XGBoost), and Support Vector Machine (SVM-RBF)—were evaluated using two feature configurations: (i) time and frequency features, and (ii) time, frequency, and chaotic features. Results demonstrate that incorporating chaotic features consistently improves recall, F1-score, and ROC-AUC across all classifiers, with a reduction in false negatives. Among the evaluated models, XGBoost achieved the best overall balance between sensitivity and precision, while SVM-RBF attained the highest recall and lowest false-negative rate, making it particularly suitable for safety-critical clinical applications. Random Forest provided competitive performance with lower computational cost. Overall, the results confirm that nonlinear chaotic EEG features provide complementary information to conventional representations and significantly improve the reliability and clinical safety of automated seizure detection systems.

KEYWORDS

Epileptic seizure detection
EEG
Chaotic dynamics
Nonlinear feature extraction
Machine learning
CHB-MIT EEG dataset

INTRODUCTION

Epilepsy is one of the most prevalent neurological disorders worldwide, affecting approximately 50 million people and accounting for nearly 5 million new diagnoses each year (World Health Organization 2024). It is characterized by recurrent, unprovoked seizures

resulting from abnormal electrical activity in the brain. Although epilepsy can occur at any age, it is most commonly diagnosed in children and older adults and can significantly affect quality of life, safety, and long-term health outcomes (Thijs *et al.* 2019).

Electroencephalography (EEG) remains the primary clinical tool for monitoring brain activity and identifying epileptic seizures because of its noninvasive nature and high temporal resolution. However, manual inspection of long-duration EEG recordings is labor-intensive and subject to inter-observer variability, motivating the development of automated seizure detection systems. Recent advances in machine learning have improved the ability to classify EEG signals, yet many existing approaches rely predominantly on

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time-domain and frequency-domain features. While these features capture amplitude and spectral characteristics of EEG signals, they may not fully represent the complex nonlinear dynamics associated with seizure generation and propagation.

The human brain is a highly nonlinear dynamical system whose electrical activity exhibits complex temporal behavior. During seizure onset and evolution, EEG signals often undergo significant changes in dynamical complexity, synchronization, and stability. Chaos theory provides a mathematical framework for characterizing such behavior through nonlinear measures including Lyapunov exponents, entropy-based metrics, and fractal characteristics. These measures can reveal dynamical information that is not captured by conventional linear analysis and may therefore improve seizure detection performance.

Motivated by these observations, this study proposes an EEG-based seizure detection framework that integrates time-domain, frequency-domain, and chaotic features extracted from multi-channel EEG recordings. The proposed approach employs a comprehensive preprocessing pipeline, feature engineering strategy, and machine-learning classifiers to evaluate the contribution of nonlinear dynamics to seizure detection performance using the CHB-MIT EEG database. The effectiveness of Random Forest, Extreme Gradient Boosting, and Support Vector Machine classifiers is investigated with and without chaotic features. The results demonstrate that incorporating nonlinear dynamical information improves detection accuracy, robustness, and sensitivity to seizure events.

BACKGROUND

This section presents the fundamental concepts underlying the proposed seizure detection framework, including epilepsy, EEG acquisition, EEG frequency bands, and chaos theory. These concepts provide the foundation for the feature extraction and machine-learning methods employed in this study.

Epilepsy

Epilepsy is a neurological disorder characterized by recurrent seizures caused by abnormal electrical activity in the brain (Fisher *et al.* 2005, 2017; Beniczky *et al.* 2025). According to the International League Against Epilepsy, seizures are broadly classified as focal or generalized depending on their origin and propagation within the brain. Focal seizures arise from a single hemisphere and may affect motor, sensory, autonomic, or cognitive functions, whereas generalized seizures involve bilaterally distributed neural networks and include several subtypes with distinct clinical manifestations. Seizure activity is commonly divided into four phases: preictal, ictal, postictal, and interictal (Usman *et al.* 2021). Understanding these phases is important for the development of automated EEG-based seizure detection systems.

EEG Electrode Placement

EEG recordings are commonly acquired using the International 10–20 System, a standardized method for positioning scalp electrodes based on anatomical landmarks (Tatum 2021). Electrodes are labeled according to the associated brain region, including frontopolar (Fp), frontal (F), central (C), temporal (T), parietal (P), and occipital (O). Odd and even numbers denote locations in the left and right hemispheres, respectively, while “z” indicates a midline position. This standardized arrangement facilitates consistent acquisition and analysis of brain activity across subjects (Beniczky and Schomer 2020).

Characteristics of EEG Signals

EEG signals are commonly analyzed according to five frequency bands: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (>30 Hz) (Attar 2022). These bands are associated with different physiological and cognitive states and provide valuable information for characterizing normal and pathological brain activity. Changes in the distribution and power of these frequency components are frequently observed during epileptic seizures, making them important features for automated detection systems.

Chaos Theory and Nonlinear EEG Analysis

Chaos theory studies (Mashuri *et al.* 2024; Fernández-Díaz 2023) deterministic nonlinear dynamical systems that exhibit complex and seemingly unpredictable behavior despite being governed by well-defined rules. A key characteristic of chaotic systems is their sensitivity to initial conditions, where small variations in the initial state can produce substantially different outcomes over time.

The brain is widely recognized as a complex nonlinear dynamical system, and EEG signals often exhibit chaotic characteristics. Consequently, nonlinear analysis has become increasingly important for characterizing seizure-related changes in brain activity. Measures such as the Largest Lyapunov Exponent, entropy metrics, and fractal dimensions can quantify signal complexity, predictability, and dynamical evolution beyond what is possible with conventional time-domain and frequency-domain analysis. These properties make chaos-based features particularly attractive for improving the detection and characterization of epileptic seizures.

RELATED WORK

Recent advances in electroencephalographic-based seizure detection have increasingly focused on combining nonlinear dynamics, machine learning, and deep learning techniques to improve detection accuracy and robustness. A dominant trend in recent literature is the integration of temporal, spectral, and chaotic feature domains to better capture the complex and nonstationary behavior of epileptic brain activity.

Several studies have demonstrated the effectiveness of nonlinear and chaos-based features for seizure detection. Measures such as the Largest Lyapunov Exponent, fractal dimensions, entropy metrics, and synchronization indices have been shown to effectively characterize changes in signal complexity associated with seizure onset and progression. Egorova *et al.* (Egorova *et al.* 2024) employed chaos-based features including Lyapunov exponents and fractal dimensions within an ensemble-learning framework and reported accuracies around 98%. Buchanna *et al.* (Buchanna *et al.* 2022) demonstrated that fractal metrics and the Hurst exponent provide strong predictive capability for seizure state classification. Similarly, Buchanna *et al.* (Buchanna *et al.* 2022) showed that epileptic EEG signals can be effectively identified using nonlinear and fractal-based features such as the Hurst exponent and adaptive fractal analysis. By combining these features with gradient-boosted decision trees and support vector regression, they reported a classification accuracy of 98.23%. Their results demonstrated that nonlinear feature extraction significantly improves EEG-based seizure detection performance. Mahgoub and Qaraqe (Mahgoub and Qaraqe 2023) further showed that synchronization-based chaotic measures can achieve high sensitivity even with relatively simple machine-learning models.

In parallel, deep learning-based approaches have achieved state-of-the-art performance in seizure detection tasks. Huang *et al.*

(Huang *et al.* 2025) proposed a hybrid convolutional and bidirectional long short-term memory network with attention mechanisms, achieving high accuracy and reduced false-alarm rates on benchmark datasets. Aravind *et al.* (KR *et al.* 2023) integrated spectral and statistical features with bidirectional long short-term memory networks to capture long-range temporal dependencies in electroencephalographic signals. Review studies such as Tan *et al.* (Tan *et al.* 2025) consistently report that hybrid models combining nonlinear features with deep learning architectures outperform traditional single-domain approaches.

Entropy-based and hybrid nonlinear approaches have also shown strong performance. Sikarwar *et al.* (Sikarwar *et al.* 2025) and Jing *et al.* (Jing *et al.* 2025) incorporated entropy measures within deep learning frameworks and reported accuracies exceeding 98%, along with improved real-time performance and reduced false-alarm rates. These results highlight the importance of signal complexity measures in enhancing seizure detection systems.

Despite recent advances, several limitations remain in current seizure detection research. Deep learning models often require large labeled datasets, high computational resources, and extensive parameter tuning, which limits their applicability in real-time or resource-constrained environments. Conversely, chaos-based approaches are computationally efficient and interpretable, but they are frequently evaluated in isolation without systematic integration with conventional time-domain and frequency-domain features. As a result, the combined contribution of temporal, spectral, and nonlinear chaotic information within a unified machine-learning framework remains insufficiently explored.

Current seizure detection research also faces broader challenges related to data variability and clinical deployment. Electroencephalographic signals vary significantly across patients, electrode placements, and seizure types, reducing model generalization capability across individuals. Physiological variability and inconsistent recording conditions further affect the stability of both traditional and deep learning models. In addition, class imbalance remains a persistent issue that can bias classification performance. Many studies also highlight the high computational cost of complex architectures and emphasize the need for low-latency, energy-efficient systems suitable for continuous monitoring. Although nonlinear and chaotic features improve robustness, their validation is often limited to small or controlled datasets, restricting clinical applicability.

To address these gaps, the present study investigates the integration of time-domain, frequency-domain, and nonlinear chaotic features extracted from multi-channel electroencephalographic signals. The proposed framework evaluates their impact on the performance of Random Forest, Extreme Gradient Boosting, and Support Vector Machine classifiers using the CHB-MIT dataset, providing a systematic assessment of the contribution of nonlinear dynamics to automated seizure detection.

MATERIALS AND METHODOLOGY

System Architecture

The system architecture, shown in Figure 3.1, follows a structured pipeline: EEG Datasets, Signal Preprocessing, Feature Extraction, Feature Reduction (PCA), Classifier, and Evaluation.

Multichannel EEG recordings with annotations are first preprocessed using bandpass filtering, wavelet denoising, segmentation, and artifact removal to improve signal quality. The signals are then divided into short time windows, each assigned a corresponding label.

For each window, features are extracted from three domains: time, frequency, and chaos/non-linear. These features capture statistical properties, spectral content, and dynamical complexity of EEG signals.

The resulting feature vectors are standardized and reduced using Principal Component Analysis (PCA) to obtain compact representations. These are then classified using Random Forest, XGBoost, and Support Vector Machine (SVM) to produce seizure detection outputs.

Model performance is evaluated using standard metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis.

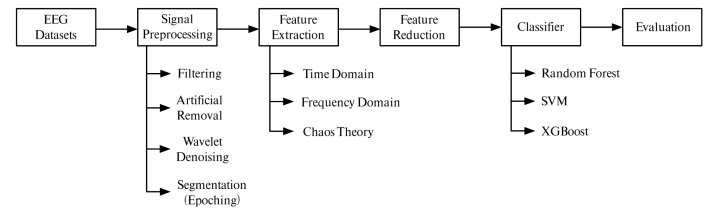


Figure 1 EEG-based machine learning pipeline with preprocessing, feature extraction, and classification stages.

Dataset and Channel Selection

The dataset used in this study is the publicly available CHB-MIT Scalp EEG Database (version 1.0.0) from PhysioNet. It contains continuous scalp EEG recordings from pediatric subjects with intractable epilepsy collected at the Children’s Hospital Boston. The database includes recordings from 22 subjects, with annotations indicating seizure onset and offset times.

All signals were acquired using the international 10–20 electrode placement system at a sampling rate of 256 Hz with 16-bit resolution. The dataset includes recordings from multiple patients, with varying recording durations, demographics, and seizure counts (Guttag 2010).

For this study, data from three patients (Patient 02, Patient 05, and Patient 06) were selected. Only EEG recordings containing seizure events (EDF files) were included in the analysis, as shown in Table 1.

Although the CHB-MIT dataset provides 23 scalp EEG channels, only a subset of channels was used in this study. A targeted channel-selection strategy was adopted to focus on brain regions most relevant to seizure activity. Previous studies have shown that frontal and temporal regions contain highly discriminative patterns associated with seizure onset and propagation, particularly in pediatric focal epilepsy.

Based on this evidence, six bipolar channels were selected: FP1–F7, F7–T7, FP1–F3, F3–C3, C3–P3, and P3–O1. These channels cover frontal, temporal, fronto-central, central-parietal, and parietal-occipital regions, providing a compact yet physiologically meaningful representation of seizure-related brain activity.

Using a reduced channel set decreases computational complexity, noise, and feature redundancy while preserving essential seizure information. Previous studies have reported that reduced-channel EEG systems can achieve performance comparable to, or better than, full-montage approaches when seizure patterns originate from localized brain regions (Acharya *et al.* 2015). Consequently, all preprocessing, feature extraction, dimensionality reduction, and classification procedures were performed using this six-channel subset.

■ **Table 1 CHB-MIT EEG dataset: patient demographics, recording duration, and seizure counts.**

Patient #	Recording Duration (hours)	No. of Seizures	Gender	Age (years)
1	38.6	7	F	11
2	35.3	3	M	11
3	38	7	F	14
4	132	4	M	22
5	39	5	F	7
6	66.73	10	F	1.5
7	67.05	3	F	14.5
8	20	5	M	3.5
9	63	4	F	10
10	50	7	M	3
11	34.8	3	F	12
12	24.7	40	F	2
13	33	12	F	3
14	26	7	F	9
15	40	20	M	16
16	19	10	F	7
17	21	3	F	12
18	36	6	F	18
19	30	3	F	19
20	26.6	8	F	6
21	32.8	4	F	13
22	31	3	F	9
23	26.6	7	F	6

Signal Preprocessing

EEG signals are highly susceptible to baseline drift, power-line interference, sensor noise, and physiological artifacts. Therefore, a preprocessing pipeline was applied to enhance signal quality while preserving seizure-related information. The preprocessing stages consisted of bandpass filtering, notch filtering, wavelet-based denoising, and artifact removal. This sequence produced clean and consistent EEG signals for subsequent segmentation and feature extraction.

Bandpass Filtering A finite impulse response (FIR) bandpass filter with cutoff frequencies of 0.5–40 Hz was applied to the continuous EEG recordings. The lower cutoff removes baseline drift while preserving delta activity, whereas the upper cutoff suppresses high-frequency noise and muscle artifacts while retaining clinically relevant EEG rhythms (Tran *et al.* 2022). The filtering operation is defined as:

$$y[n] = \sum_{k=0}^M h[k]x[n-k] \quad (1)$$

where $x[n]$ is the input EEG signal, $h[k]$ represents the FIR filter coefficients, $y[n]$ is the filtered signal, and M is the filter order.

Notch Filtering To suppress power-line interference, a 60 Hz notch filter was applied, consistent with the electrical frequency used in the United States where the CHB-MIT dataset was acquired. The notch filter transfer function is given by:

$$H(s) = \frac{s^2 + \omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2} \quad (2)$$

where $\omega_0 = 2\pi f_0$ is the notch center frequency and Q is the quality factor. A value of $Q = 30$ was selected to provide effective attenuation of power-line noise while minimizing distortion of adjacent EEG frequencies.

Wavelet Denoising Wavelet denoising was employed to reduce broadband noise while preserving transient EEG characteristics. The Daubechies-4 (db4) wavelet was selected due to its effectiveness in EEG analysis. The denoising process involved multi-level decomposition using the Daubechies-4 (db4) wavelet, followed by noise estimation from the finest-scale detail coefficients using the median absolute deviation (MAD) method. Soft-thresholding was applied using the universal threshold

$$\lambda = \sigma\sqrt{2 \ln N} \quad (3)$$

where σ represents the estimated noise level and N is the signal length. The denoised signal was subsequently reconstructed using the inverse discrete wavelet transform (IDWT).

Artifact Removal Residual artifacts were mitigated using statistical clipping. For each EEG channel, samples exceeding $\pm 5\sigma$ from the channel mean were clipped to the nearest boundary:

$$x_{\text{clip}} = \begin{cases} \mu + 5\sigma, & x > \mu + 5\sigma \\ \mu - 5\sigma, & x < \mu - 5\sigma \\ x, & \text{otherwise} \end{cases} \quad (4)$$

where μ and σ denote the channel mean and standard deviation, respectively. This approach reduces the influence of extreme artifacts such as muscle bursts, electrode pops, and abrupt movements while preserving seizure-related EEG activity.

The resulting cleaned signals were then forwarded to the segmentation and labeling stage.

Segmentation and Labeling

Following preprocessing, the continuous EEG recordings were segmented into fixed-length windows to generate uniform samples for feature extraction and classification. A window length of 4 s with 50% overlap (2 s hop size) was adopted, as this configuration provides a good balance between temporal resolution and feature

stability in seizure detection applications (Christou *et al.* 2022). For a sampling frequency f_s , the number of samples per window and hop size are given by:

$$N_w = 4f_s, \quad (5)$$

$$N_h = 2f_s, \quad (6)$$

where N_w and N_h denote the window and hop lengths, respectively. For the CHB-MIT dataset ($f_s = 256$ Hz), each window contains 1024 samples and consecutive windows overlap by 512 samples.

Segmentation was performed on the continuous preprocessed EEG signals to avoid boundary artifacts that may arise when filtering pre-segmented data. Each window was assigned a binary label based on the expert seizure annotations. A window was labeled as seizure (1) if more than 50% of its samples overlapped with an ictal interval; otherwise, it was labeled as non-seizure (0). This majority-coverage criterion minimizes ambiguously labeled windows and ensures that each segment reflects the dominant physiological state.

The use of overlapping windows improves sensitivity around seizure onset and offset boundaries while providing smoother temporal transitions between consecutive predictions. The resulting labeled segments were subsequently used for feature extraction and supervised machine learning.

Feature Extraction

Feature extraction was performed on each segmented EEG window to obtain a compact representation of seizure-related activity. To capture complementary characteristics of the EEG signal, features were extracted from three domains: time, frequency, and nonlinear (chaos). This multi-domain approach enables the characterization of amplitude variations, spectral changes, and dynamical complexity associated with epileptic seizures Mohammed *et al.* (2021).

Time-Domain Features Time-domain features describe the temporal behavior of the EEG waveform and are sensitive to amplitude fluctuations, transient events, and signal irregularities. Five features were extracted: Line Length (LL), Variance, and the three Hjorth parameters (Activity, Mobility, and Complexity).

Line Length measures the cumulative absolute difference between consecutive samples and reflects signal roughness and local complexity. During pre-ictal and ictal periods, the EEG often exhibits more rapid fluctuations and spikes, which cause LL to increase compared to inter-ictal baseline. Several seizure detection studies use LL as a robust, low-cost feature for early changes in excitability

$$LL = \sum_{n=1}^{N-1} |x(n+1) - x(n)| \quad (7)$$

Variance quantifies signal power and amplitude dispersion. Ictal EEG shows large amplitude rhythmic discharges, and pre-ictal segments can exhibit increased variance compared to resting activity.

$$Var = \frac{1}{N} \sum_{n=1}^N (x(n) - \mu)^2 \quad (8)$$

where μ is the signal mean. Hjorth Activity is equivalent to the signal variance. Activity typically increases in seizure and pre-ictal segments, reflecting the enhanced intensity of cortical activity. Combining Activity with Mobility and Complexity gives a richer

description than amplitude alone. Hjorth Mobility and Complexity characterize the dominant frequency content and waveform irregularity, respectively.

$$Mobility = \sqrt{\frac{Var(\Delta x)}{Var(x)}}; \quad \Delta x = x(n) - x(n-1) \quad (9)$$

Mobility approximates the mean frequency of the signal in the time domain. Higher Hjorth Mobility indicates faster oscillatory activity and a greater rate of signal variation. Changes in dominant EEG frequencies during seizure onset affect Hjorth Mobility, making it a useful feature for differentiating seizure and non-seizure states.

$$Complexity = \frac{Mobility(\Delta x)}{Mobility(x)} \quad (10)$$

Hjorth Complexity quantifies the degree of waveform irregularity. Values close to 1 indicate a nearly sinusoidal signal, while higher values reflect more complex, multi-frequency behavior. During seizure activity, EEG signals often become more rhythmic and stereotyped, leading to a reduction in Hjorth Complexity compared to baseline. The combined use of Mobility and Complexity provides a robust characterization of EEG dynamics and is widely used in epilepsy analysis.

These features are widely used in EEG analysis because seizure activity often produces increased signal power, rapid fluctuations, and changes in waveform regularity.

Frequency-Domain Features Frequency-domain features characterize the distribution of signal power across EEG frequency bands and capture seizure-related spectral changes. Power spectral density (PSD) was estimated for each EEG segment, from which six features were extracted: Delta Power (0.5–4 Hz), Alpha Power (8–13 Hz), Beta Power (13–30 Hz), Spectral Entropy, Delta-to-Alpha Ratio (DAR), and Theta-to-Alpha Ratio (TAR). Band powers were computed by integrating the PSD within the corresponding frequency ranges:

$$\begin{aligned} P_\delta &= \sum_{f=0.5}^4 P(f) && \text{(Delta power, 0.5–4 Hz)} \\ P_\alpha &= \sum_{f=8}^{13} P(f) && \text{(Alpha power, 8–13 Hz)} \\ P_\beta &= \sum_{f=13}^{30} P(f) && \text{(Beta power, 13–30 Hz)} \end{aligned} \quad (11)$$

Delta activity (0.5–4 Hz) is associated with slow-wave processes, deep sleep, and pathological slowing, and is typically increased during ictal and peri-ictal states, reflecting hypersynchronization and abnormal cortical dynamics, particularly in focal seizures. Alpha activity (8–13 Hz), dominant during relaxed wakefulness and generated mainly by thalamo-cortical networks, is often suppressed during seizures and may also decrease in pre-ictal periods, indicating disruption of normal thalamo-cortical function. Beta activity (13–30 Hz), linked to alertness, motor activity, and active cortical processing, exhibits variable changes during seizure onset depending on seizure type and brain region, and is therefore commonly used as a discriminative feature in epileptic EEG analysis.

Spectral Entropy was calculated as

$$H_{spec} = - \sum p(f) \log p(f) \quad (12)$$

where

$$p(f) = \frac{P(f)}{\sum P(f)} \quad (13)$$

is the normalized spectrum. Spectral entropy quantifies the irregularity or flatness of the EEG power spectrum. High spectral entropy indicates a broadly distributed spectrum with energy spread across multiple frequencies, reflecting complex activity, whereas low spectral entropy corresponds to power concentration within a narrow frequency range, indicating more rhythmic and organized dynamics. During seizures, ictal EEG typically exhibits dominant rhythmic activity, leading to a reduction in spectral entropy, while pre-ictal periods may show a gradual decrease reflecting emerging spectral organization.

The DAR and TAR features are defined as

$$DAR = \frac{P_\delta}{P_\alpha} \quad (14)$$

$$TAR = \frac{P_\theta}{P_\alpha} \quad (15)$$

The DAR combines slow-wave (delta) and resting-state (alpha) activity, where higher values indicate relative slowing due to delta dominance over alpha. During seizure activity, delta power typically increases while alpha decreases, resulting in an elevated DAR that serves as a clinically interpretable biomarker of pathological slowing and seizure-related dynamics. Similarly, the TAR quantifies the relative contribution of theta versus alpha activity, with increased values indicating enhanced slow-wave activity and reduced alpha power. In epilepsy studies, TAR is often used as a complementary indicator to DAR for identifying abnormal or pre-ictal states compared to normal alpha-dominant wakefulness. These features capture the spectral slowing, rhythm reorganization, and power redistribution commonly observed during seizure onset and progression.

Nonlinear (Chaos) Features Nonlinear features quantify the complexity, irregularity, and dynamical structure of EEG signals. Unlike conventional time- and frequency-domain measures, nonlinear descriptors capture the underlying behavior of the brain as a complex dynamical system. Since epileptic seizures are often characterized by a transition from highly complex neural activity to more synchronized and low-dimensional dynamics, nonlinear measures provide valuable discriminatory information for distinguishing inter-ictal, pre-ictal, and ictal states [Rodriguez-Bermudez and Garcia-Laencina \(2015\)](#); [Kargarnovin et al. \(2023\)](#); [Jorgsson \(2023\)](#).

Six nonlinear features were extracted from each EEG segment: Sample Entropy (SampEn), Approximate Entropy (ApEn), Correlation Dimension (CD), Detrended Fluctuation Analysis (DFA), Higuchi Fractal Dimension (HFD), and Largest Lyapunov Exponent (LLE). SampEn and ApEn quantify signal regularity by estimating the likelihood that similar patterns remain similar over time, with lower values indicating more predictable and less complex dynamics [Olbrys and Majewska \(2022\)](#); [Montesinos et al. \(2018\)](#); [Zhang and He \(2018\)](#). Correlation Dimension estimates the dimensionality of the reconstructed phase space and is commonly used to characterize the complexity of nonlinear systems [Brari and Belghith \(2021\)](#). Detrended Fluctuation Analysis evaluates long-range temporal correlations through the scaling relationship $F(s) \propto s^\alpha$ where $F(s)$ is the fluctuation function, s is the observation scale, and α is the scaling exponent that reflects correlation properties of the signal [Gaurav et al. \(2021\)](#). Higuchi Fractal Dimension measures waveform complexity directly in the time domain

by estimating the scaling behavior of signal length across multiple observation intervals [Masoumirad et al. \(2026\)](#). Finally, the Largest Lyapunov Exponent quantifies the average exponential divergence of neighboring trajectories in phase space, $d(t) \approx (0)e^{\lambda t}$ where $d(t)$ represents the separation between trajectories at time t and λ denotes the Lyapunov exponent. Positive values of λ indicate chaotic dynamics, whereas lower values suggest increased synchronization and reduced complexity [Brari and Belghith \(2022\)](#).

In this study, the Largest Lyapunov Exponent was computed using Rosenstein's algorithm, which is computationally efficient and well suited for short EEG segments [Hayashi \(2024\)](#). The final 17-feature set was selected to be compact, non-redundant, and physiologically meaningful across inter-ictal, pre-ictal, and ictal states. Time-domain features (Line Length, Variance, and Hjorth Activity, Mobility, and Complexity) characterize waveform intensity, temporal variability, and signal irregularity. Frequency-domain features (Delta, Alpha, and Beta powers, Spectral Entropy, DAR, and TAR) capture spectral reorganization and power redistribution associated with seizure activity. Nonlinear features (Sample Entropy, Approximate Entropy, Correlation Dimension, DFA, Higuchi Fractal Dimension, and Largest Lyapunov Exponent) quantify changes in signal complexity, long-range temporal correlations, and underlying dynamical behavior. Commonly overlapping or noise-sensitive features, such as RMS, skewness, kurtosis, median frequency, and gamma-band power, were excluded to improve robustness and generalization. Collectively, these complementary descriptors provide a compact multi-domain representation of EEG activity, enhancing seizure discrimination and improving cross-patient classification performance.

Feature Reduction

To reduce feature dimensionality while preserving the most informative characteristics of the EEG data, Principal Component Analysis (PCA) was applied to the extracted feature set. PCA transforms the original correlated features into a smaller set of orthogonal principal components that capture the maximum variance in the data [Bharadiya \(2023\)](#); [Salem and Hussein \(2019\)](#).

Prior to PCA, all features were standardized using z-score normalization: $x' = \frac{x - \mu}{\sigma}$ where μ and σ denote the mean and standard deviation of each feature, respectively. Standardization parameters were computed using the training data only and subsequently applied to the validation and test sets to prevent data leakage.

Let $X \in \mathbb{R}^{n \times p}$ denote the standardized feature matrix. PCA was performed by eigendecomposition of the covariance matrix: $C = \frac{1}{n-1} X^T X$ yielding eigenvectors and corresponding eigenvalues that represent the principal directions and their explained variance. The transformed feature matrix is obtained as $Z = XW_k$ where W_k contains the first k principal components and Z represents the reduced feature space.

The number of retained components was selected such that the cumulative explained variance exceeded 95%. This approach reduces feature redundancy, mitigates multicollinearity, and lowers the risk of overfitting while preserving the majority of discriminative information. PCA has been widely adopted in EEG-based classification systems due to its ability to produce compact and separable feature representations that improve computational efficiency and model generalization.

Classification

The reduced feature set obtained from PCA was used to train supervised binary classifiers for seizure and non-seizure discrimination. Three machine learning models were evaluated: Support

Vector Machine (SVM), Random Forest (RF), and Extreme Gradient Boosting (XGBoost). These classifiers were selected due to their proven effectiveness in EEG-based seizure detection and their ability to model nonlinear relationships within high-dimensional feature spaces.

Support Vector Machine A Support Vector Machine with a Radial Basis Function (RBF) kernel was employed as the primary classifier Hosseini *et al.* (2020). The RBF kernel enables nonlinear separation of EEG feature vectors in a higher-dimensional space and has demonstrated strong performance on biomedical classification tasks. To address class imbalance, class weighting was applied to penalize misclassification of seizure samples.

The RBF kernel is defined as $K(x_i, x_j) = \exp(-\gamma|x_i - x_j|^2)$ where γ controls the influence of individual training samples. The SVM determines an optimal separating hyperplane by solving

$$\min_{w, b, \xi} \frac{1}{2} |w|^2 + C \sum_{i=1}^N \xi_i \quad (16)$$

subject to

$$y_i (w^T \phi(x_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad (17)$$

where C is the regularization parameter and ξ_i are slack variables that allow soft-margin classification.

Random Forest Random Forest Jiang *et al.* (2023) is an ensemble learning method that combines multiple decision trees trained on bootstrap samples with random feature selection. Final predictions are obtained through majority voting across all trees. Random Forests are robust to noise and outliers, effectively capture nonlinear feature interactions, and provide estimates of feature importance that enhance model interpretability.

XGBoost XGBoost Wei and Mooney (2020) is a gradient boosting algorithm that sequentially constructs decision trees to minimize a regularized objective function. At iteration t , the objective is expressed as

$$\sum_i \ell(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \quad (18)$$

where $\ell(\cdot)$ denotes the classification loss function and $\Omega(f_t)$ is a regularization term controlling model complexity. XGBoost is particularly effective for structured tabular data and can efficiently handle nonlinear relationships and class imbalance.

Experimental Setup and Training Protocol The proposed framework is designed for offline window-level seizure detection from scalp EEG recordings, focusing on binary seizure versus non-seizure classification. The EEG windows were partitioned into training and testing sets using an 80/20 stratified split to preserve class distributions. To ensure reproducibility and prevent data leakage, all preprocessing, feature standardization, and PCA transformations were fitted exclusively on the training data, and the learned parameters were subsequently applied to the test set. All classifiers were trained using baseline hyperparameter settings to enable a fair comparison of their performance. Model evaluation was conducted using clinically relevant metrics, with particular emphasis on sensitivity and false-negative analysis due to the critical importance of minimizing missed seizure events.

Evaluation Protocol

The performance of the proposed seizure detection system was evaluated using a set of standard classification metrics (Accuracy, Precision, Recall (Sensitivity), Specificity, and F1-score) derived from the confusion matrix. These metrics provide complementary insights into the model's ability to correctly identify seizure and non-seizure EEG segments.

The confusion matrix consists of four outcomes: True Positives (TP), representing correctly detected seizure windows; True Negatives (TN), representing correctly identified non-seizure windows; False Positives (FP), corresponding to false seizure alarms; and False Negatives (FN), representing missed seizures. Based on these quantities, the following evaluation metrics were computed:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (19)$$

$$Precision = \frac{TP}{TP + FP} \quad (20)$$

$$Recall; (Sensitivity) = \frac{TP}{TP + FN} \quad (21)$$

$$Specificity = \frac{TN}{TN + FP} \quad (22)$$

$$F_1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (23)$$

In addition, the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) was calculated to assess the model's ability to discriminate between seizure and non-seizure classes across different decision thresholds.

Particular emphasis was placed on Recall (Sensitivity) and False Negatives due to their clinical significance. Missing a seizure event may delay diagnosis or intervention and therefore represents the most critical type of classification error. Consequently, false negatives were explicitly analyzed alongside the overall performance metrics. The confusion matrix was also examined both numerically and visually to provide a comprehensive assessment of classifier behavior and to identify potential areas for improvement.

RESULTS AND DISCUSSION

Performance was evaluated using the three prementioned classifiers under two feature configurations: (i) time and frequency features, and (ii) time, frequency, and nonlinear (chaos) features. A total of 59,671 EEG windows across six channels were analyzed.

Signal Preprocessing Results

Raw EEG recordings were substantially affected by baseline drift, broadband noise, and artifact contamination, resulting in a low signal-to-noise ratio (SNR) of 2.82 dB. The proposed preprocessing pipeline, consisting of band-pass filtering (0.5–40 Hz), 60 Hz notch filtering, wavelet-based denoising, and statistical artifact suppression, significantly improved signal quality. As shown in Figure 2, basic preprocessing increased the SNR to 24.14 dB, while the complete enhanced pipeline further increased it to 24.42 dB. Visual inspection indicates improved waveform clarity and preservation of seizure-related rhythmic activity. The corresponding power spectral density (PSD) analysis, shown in Figure 3, demonstrates effective attenuation of out-of-band noise and spectral artifacts while maintaining power within clinically relevant EEG frequency bands. These results confirm that the preprocessing framework enhances signal quality and provides a stable foundation for reliable feature extraction and seizure classification.

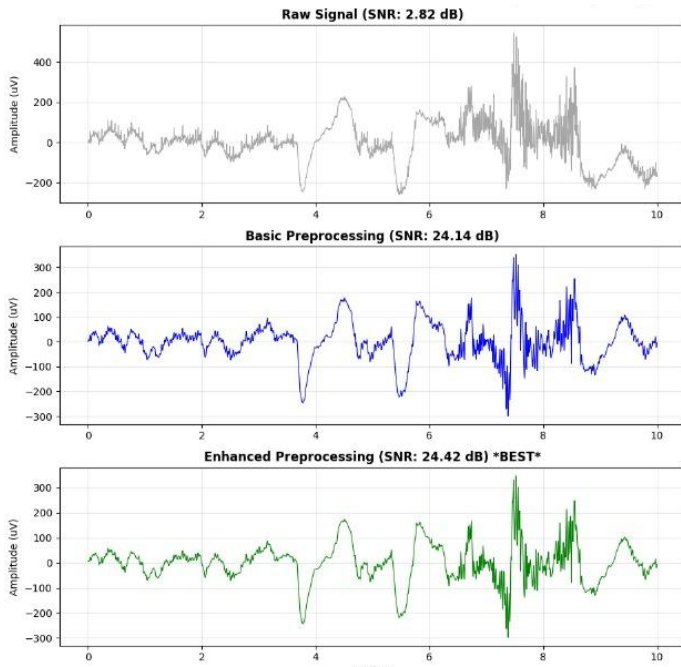


Figure 2 Preprocessing *chb02₁₆.edf*, *channel0*.

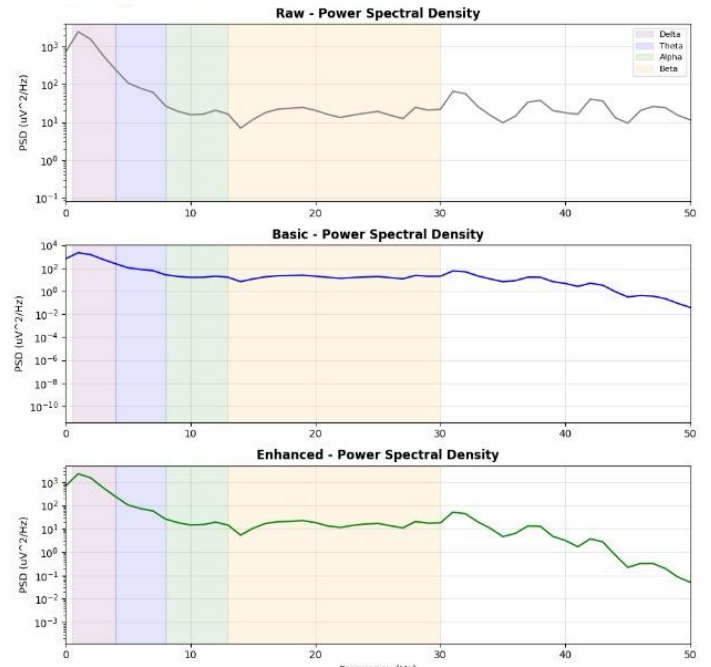


Figure 3 PSD of *chb02₁₆.edf*, *channel0*.

Table 2 PCA dimensionality reduction results.

Feature Set	Original	PCA	Variance
	Features	Components	
Time & Frequency	66	23	95.35%
Time, Frequency, & Chaos	102	29	95.03%

Feature Extraction

Time-, frequency-, and chaos-domain features were extracted from each EEG window to capture complementary characteristics of seizure activity. The final feature set comprised 66 features for the Time + Frequency configuration and 102 features for the Time + Frequency + Chaos configuration. These features describe signal amplitude variations, spectral organization, and nonlinear dynamical behavior associated with epileptic seizures.

PCA Dimensionality Reduction

Principal Component Analysis (PCA) was applied to both feature configurations to reduce dimensionality while preserving at least 95% of the total variance. As shown in Table 2, the Time + Frequency feature set (66 original features) was reduced to 23 principal components while retaining 95.35% of the variance. Similarly, the Time + Frequency + Chaos feature set (102 features) was reduced to 29 components while preserving 95.03% of the variance.

PCA substantially compressed the feature space, reducing dimensionality by 65.2% and 71.6% for the two feature configurations, respectively, while preserving the majority of discriminative information. This reduction decreases computational complexity, mitigates overfitting, and improves the efficiency of subsequent classification models.

Class Imbalance Handling

The dataset exhibits a severe class imbalance, with 47,389 non-seizure windows (99.3%) and only 347 seizure windows (0.7%). Such imbalance can bias learning algorithms toward the majority class and significantly reduce sensitivity to seizure events.

To address this issue, the Synthetic Minority Over-sampling Technique (SMOTE) [Pradipta et al. \(2021\)](#) was applied on the training data to generate synthetic seizure samples and balance the class distribution. This approach produces new minority class instances by interpolating between existing seizure samples in feature space, rather than duplicating data.

After oversampling, the training dataset was balanced, resulting in 94,778 samples with an equal distribution of seizure and non-seizure classes (50% each). This balancing strategy improves classifier sensitivity (Recall) and enhances robustness by reducing bias toward the dominant class, leading to more stable performance across all evaluated models.

Random Forest Classifier Results

Table 3 summarizes the performance of the Random Forest classifier under the two feature configurations.

The incorporation of chaos-based nonlinear features, including the Largest Lyapunov Exponent and other complexity measures, improves the performance of the Random Forest classifier across all evaluation metrics. As shown in Figure 4, Recall increases from 75.86% to 80.46% (+6.1%), indicating a higher proportion of correctly detected seizure events. Precision also improves from 90.41% to 93.33% (+3.2%), while the F1-score rises from 82.50% to 86.42% (+4.8%), demonstrating a better balance between sensitivity and false-alarm reduction. The ROC-AUC increases from 0.9828 to 0.9944 (+1.2%), confirming improved discrimination between seizure and non-seizure windows.

The confusion matrices in Figure 4 further illustrate these gains. The number of true positive detections increases from 66 to 70, while false negatives decrease from 21 to 17, representing a 19.0% reduction in missed seizures. Simultaneously, false positives de-

Table 3 Random Forest performance comparison across feature sets.

Metric	Time & Freq.	& Chaos	Improvement
Accuracy	99.77%	99.82%	+0.05%
Precision	90.41%	93.33%	+3.22%
Recall	75.86%	80.46%	+6.10%
F1-Score	82.50%	86.42%	+4.80%
ROC-AUC	0.9828	0.9944	+1.16%
False Negatives	21	17	-19.0%

crease from 7 to 5, indicating improved specificity without sacrificing sensitivity. These results demonstrate that chaos-based features provide complementary information beyond conventional time- and frequency-domain descriptors, enabling the Random Forest model to better capture seizure-related dynamics and improve clinically relevant detection performance.

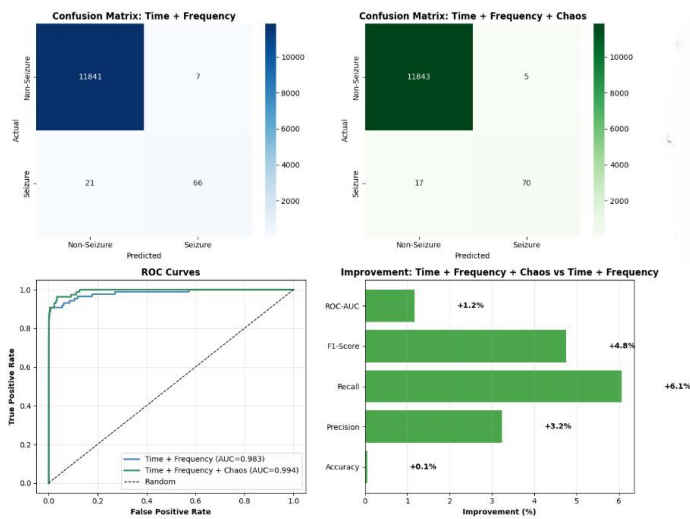


Figure 4 Random Forest performance comparison using Time+Frequency features and Time+Frequency+Chaos features, showing confusion matrices, ROC curves, and metric improvements.

XGBoost Classifier Results

Table 4 summarizes the performance of the XGBoost classifier under the two feature configurations.

The addition of chaos-based nonlinear features significantly enhances the performance of the XGBoost classifier across all evaluation metrics. As shown in Figure 5, Precision, Recall, and F1-score each increase from 82.76% to 88.51%, corresponding to a relative improvement of approximately 6.9%. Accuracy also increases slightly from 99.75% to 99.83%, while the ROC-AUC improves from 0.9887 to 0.9925, indicating enhanced discrimination between seizure and non-seizure classes.

The confusion matrices further demonstrate the benefit of incorporating nonlinear dynamics. The number of correctly detected seizure windows (true positives) increases from 72 to 77, while

Table 4 XGBoost performance comparison across feature sets.

Metric	Time & Freq.	& Chaos	Relative Gain
Accuracy	99.75%	99.83%	+0.08%
Precision	82.76%	88.51%	+5.75%
Recall	82.76%	88.51%	+5.75%
F1-Score	82.76%	88.51%	+5.75%
ROC-AUC	0.9887	0.9925	+0.38%
False Negatives	15	10	-33.3%

false negatives decrease from 15 to 10, representing a 33.3% reduction in missed seizures. At the same time, false positives are reduced from 15 to 10, leading to improved precision without compromising sensitivity. The ROC curves show a consistent improvement across operating thresholds, confirming the additional discriminative value provided by chaos-based features.

Overall, XGBoost achieves the most balanced performance among the evaluated classifiers, combining high sensitivity, strong precision, and low false-negative rates. These results suggest that nonlinear dynamical descriptors provide complementary information to conventional time- and frequency-domain features, enabling XGBoost to more effectively characterize seizure-related EEG patterns.

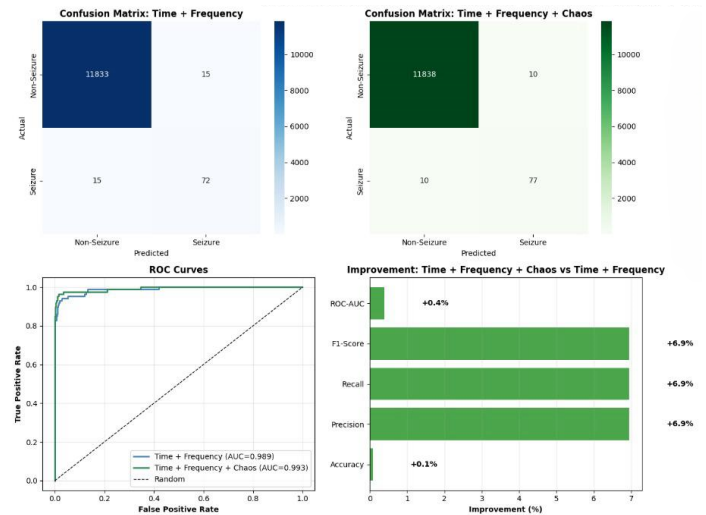


Figure 5 XGBoost performance comparison using Time+Frequency features and Time+Frequency+Chaos features, showing confusion matrices, ROC curves, and metric improvements.

SVM-RBF Classifier Results

Table 5 summarizes the performance of the SVM classifier under the two feature configurations.

The inclusion of chaos-based nonlinear features produces the largest relative performance improvement for the SVM-RBF classifier. As shown in Figure 6, Precision increases from 51.33% to 74.53%, representing a substantial gain of 45.2% and a significant

■ **Table 5 SVM-RBF performance comparison across feature sets.**

Metric	Time & Freq.	& Chaos	Relative Gain
Accuracy	99.30%	99.71%	+0.41%
Precision	51.33%	74.53%	+45.20%
Recall	88.51%	90.80%	+2.29%
F1-Score	64.98%	81.87%	+26.89%
ROC-AUC	0.9929	0.9977	+0.48%
False Negatives	10	9	-10%

reduction in false alarms. Recall also improves from 88.51% to 90.80% (+2.6%), indicating enhanced seizure detection sensitivity. Consequently, the F1-score rises from 64.98% to 81.87% (+26.0%), reflecting a markedly better balance between sensitivity and precision.

The confusion matrices further highlight the benefits of incorporating chaos-based features. The number of true positives increases from 77 to 79, while false negatives decrease from 10 to 8, corresponding to a 20% reduction in missed seizures. More notably, false positives decrease substantially from 73 to 27, explaining the large improvement in precision and overall classification reliability. The ROC-AUC increases from 0.9929 to 0.9977 (+0.5%), indicating excellent separability between seizure and non-seizure classes across decision thresholds.

Among all evaluated classifiers, SVM-RBF achieves the highest sensitivity (90.80%), the lowest false-negative count (8), and the highest ROC-AUC (0.9977) when chaos-based features are included. These results demonstrate that nonlinear dynamical descriptors provide highly discriminative information that complements conventional time- and frequency-domain features, making SVM-RBF particularly suitable for safety-critical seizure monitoring applications where minimizing missed detections is a primary objective.

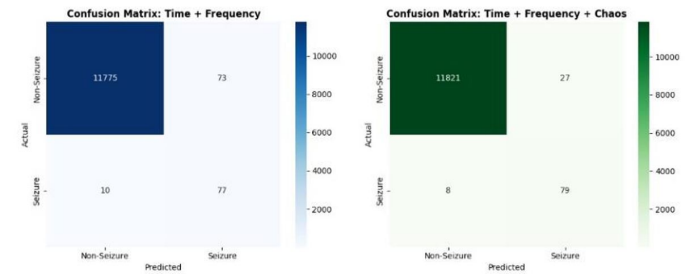


Figure 6 SVM-RBF performance comparison using Time+Frequency features and Time+Frequency+Chaos features, showing confusion matrices, ROC curves, and metric improvements.

Classifier Comparison

Table 6 summarizes the performance of all classifiers using the full feature set (Time + Frequency + Chaos). Overall, all models achieve high accuracy; however, clear differences emerge in sensitivity, false negative rate, and class balance.

■ **Table 6 Performance comparison of classifiers using Time + Frequency + Chaos features.**

Classifier	Accuracy	Precision	Recall	F1-Score	ROC-AUC
RF	99.82%	93.33%	80.46%	86.42%	0.9944
XGBoost	99.83%	88.51%	88.51%	88.51%	0.9925
SVM-RBF	99.71%	74.53%	90.80%	81.87%	0.9977

Despite comparable overall accuracy across models, their behavior differs significantly in terms of error distribution. XGBoost achieves the most balanced performance between precision and recall, resulting in strong overall robustness. Random Forest exhibits high precision but lower sensitivity, indicating a tendency toward missing seizure events. In contrast, SVM-RBF achieves the highest recall and lowest false-negative tendency, making it the most sensitive model for seizure detection, although at the cost of reduced precision.

These results highlight a fundamental trade-off between sensitivity and false alarm control across classifiers. From a clinical perspective, models with higher recall (XGBoost and SVM-RBF) are more favorable, as minimizing missed seizures is critical for patient safety.

Benchmarks and Result Validity

The reported performance on the CHB-MIT dataset is consistent with prior studies, which typically achieve accuracies between 97% and 99.5% and sensitivity-related metrics between 85% and 98% depending on feature engineering and preprocessing strategies [Liu et al. \(2023\)](#); [Zougagh et al. \(2025\)](#); [Ali et al. \(2024\)](#). The results obtained in this work fall within this established range, confirming methodological validity and comparability with existing literature. While previous studies report typical improvements of 5–15% when incorporating nonlinear features, a larger gain was observed for the SVM-RBF model due to a substantial reduction in false positives, highlighting the discriminative strength of the proposed nonlinear feature set.

CONCLUSION

This study presented a complete EEG-based seizure detection framework based on multi-channel scalp EEG analysis, enhanced preprocessing, optimized feature engineering, and supervised machine learning classifiers including Random Forest, XGBoost, and SVM with RBF kernel. The CHB-MIT dataset was processed using a four-stage pipeline comprising band-pass filtering, notch filtering, wavelet denoising, and artifact suppression, resulting in a substantial improvement in signal-to-noise ratio and clearer seizure morphology.

A comprehensive feature extraction strategy was applied across three domains: time, frequency, and nonlinear chaotic dynamics. A compact and physiologically informed feature set was constructed, followed by PCA-based dimensionality reduction to improve computational efficiency and reduce redundancy. Two experimental configurations were evaluated: (i) Time + Frequency features and (ii) Time + Frequency + Chaos features. Across all classifiers, the inclusion of chaotic features consistently improved performance, particularly in Recall, F1-score, ROC-AUC, and false-negative reduction.

Experimental results demonstrate that nonlinear chaotic features, especially the Largest Lyapunov Exponent, significantly enhance seizure detection performance. In clinical epilepsy monitoring, Recall (Sensitivity) is the most critical metric, as missed seizures may lead to severe consequences including injury and potential SUDEP. Therefore, minimizing false negatives is prioritized over marginal gains in overall accuracy. Among all models, SVM-RBF achieved the highest clinical safety performance with a Recall of 90.80% and only eight missed seizures, along with the highest ROC-AUC (0.9977). The addition of chaos features also improved its precision by 45.2%, significantly reducing false alarms while maintaining strong sensitivity, making it highly suitable for high-risk environments such as ICU and continuous home monitoring systems.

XGBoost achieved the most balanced overall performance, with Recall and F1-score both reaching 88.51% and a ROC-AUC of 0.9925. It also demonstrated a substantial reduction in false negatives, indicating strong reliability for general-purpose seizure detection systems requiring both stability and clinical safety.

Random Forest benefited from chaotic features as well, improving Recall from 75.86% to 80.46% and reducing false negatives from 21 to 17, while achieving the highest precision (93.33%). Its computational efficiency and interpretability make it suitable for resource-constrained or explainable AI applications.

In general, the results confirm that epileptic seizures exhibit a nonlinear dynamical behavior that cannot be fully captured by traditional time-frequency analysis alone. The integration of chaos-based features provides consistent improvements across all classifiers, enhancing robustness, generalization, and most importantly, clinical safety in seizure detection systems.

Availability of data and material

Not applicable.

Conflicts of interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

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